Kaan: On-Device Acoustic Screening for Stored-Grain Pests under Phone-Mic Domain Shift

Anonymous Author(s)

Abstract

Post-harvest insect damage is a major food-security and climate-relevant loss pathway in stored grain. We present **Kaan**, an open, offline smartphone system that classifies contact acoustics into clean grain versus three primary stored-rice pests using a compact INT8 CNN. Rather than chasing laboratory accuracy alone, we emphasize deployment constraints: leakage-aware evaluation, multi-approach bake-offs, teacher distillation into a phone-sized model, temperature calibration with selective abstention, and a *robustness ladder* that stress-tests phone-like degradations without new field recordings. On IRRI acoustic data with ambient clean windows, a distilled production model reaches 97.2% validation accuracy; multi-seed bake-offs show classical and deep models both exceed a cited 84.51% reference under our protocol. The robustness ladder reveals sharp drops under additive noise and telephone band-limiting—a limitation we treat as a first-class result for climate-adaptation tooling that must survive rural handsets. Code and models are released under Apache-2.0.

1 Climate impact pathway

Reducing avoidable post-harvest loss lowers pressure to expand production and associated emissions, while improving food availability for smallholders IGMRI (2015); World Bank and FAO. Acoustic screening can catch internal feeders before visual symptoms, enabling earlier aeration, sunning, or fumigation decisions. Kaan targets *adaptation* (protecting stored food under climate stress on storage conditions) and indirect *mitigation* (less waste). We deliberately optimize for offline phones: no cloud inference, multilingual advisories, and models small enough for browser/ONNX and TFLite.

2 Problem and system

Classes: clean, rice weevil (*Sitophilus oryzae*), lesser grain borer (*Rhyzopertha dominica*), red flour beetle (*Tribolium castaneum*). **Pipeline:** 10 s @ 16 kHz mono $\rightarrow$ 128×128 mel $\rightarrow$ INT8 CNN $\rightarrow$ confidence gate. Pest WAVs come from the IRRI Rice Acoustic Sensor dataset Balingbing et al. (2024); clean uses Speech Commands ambient windows rather than synthetic silence. Splits are stratified at *file* level after byte-deduplication to avoid train/val twins.

3 Methods

Bake-off. Eleven approaches (mel-CNNs, 1D CNN, YAMNet probe, SVM/MLP/GBDT/RF/ExtraTrees/kNN/logreg) share one leakage-aware split; we report multi-seed means, bootstrap CIs, and McNemar tests.

Preprint.

Table 1: Multi-seed bake-off accuracy (mean±std; seeds 42/43/44).

Approach	Acc mean±std	Seeds $> 84.51\%$
GBDT	95.36 ± 1.50	3/3
CNN deep	95.15 ± 0.48	3/3
ExtraTrees	94.94 ± 1.10	3/3
SVM-RBF	94.73 ± 1.20	3/3
YAMNet probe	85.65 ± 1.83	2/3
CNN-1D	64.77 ± 20.7	1/3

Table 2: Robustness ladder on seed-42 baseline (clean $= 97.15\%$).

Rung	Acc
Muffle / reverb / compress / gain	92–97%
Clip	87.7%
Phone band-pass	52.2%
SNR 20 dB	30.4%
SNR ≤ 10 dB / phone combos	$\sim 7.6\%$

Distillation. Soft labels from GBDT + ExtraTrees + deep CNN train a deep student for deployment (TFLite $\sim$333 KB; ONNX for web).

Robustness ladder. Without field mics, we replay validation audio under pink SNR, phone band-pass, muffling, compression, clipping, reverb, gain/AGC, and combinations—reporting accuracy per rung.

Calibration. Temperature scaling fitted on a validation slice; abstain curves trade coverage for accuracy.

Confusion-focused heads. Cost-sensitive upweighting and a fine-tuned weevil$\leftrightarrow$borer specialist on the baseline encoder with *strict* fusion (override only when top-2 are that pair and margin is small; gate swept on val).

SSL. SimCLR-style SpecAugment views on IRRI+ambient mels, then supervised fine-tune.

4 Results

Table 1 summarizes the multi-seed bake-off (reference line: cited 84.51% Balingbing et al. (2024) under *our* protocol, not a locked reimplementation). Distillation (seed 42) yields **97.15%** val accuracy (macro-F1 0.977) versus 95.57% hard-only deep and 95.89% teacher ensemble. SSL fine-tune reaches 97.78% on the same seed. Calibration reduces ECE (e.g., $0.044 \rightarrow 0.030$ at T=0.5 on seed 42).

Table 2 is the deployment-facing result: laboratory mel-CNNs survive mild muffling but **fail under phone-band and noise**—exactly the shift rural handsets induce. From-scratch hierarchical cascades collapse; fine-tuned specialists reach $\sim$97% on the pair subset alone, and strict gating aims to preserve baseline overall accuracy (multi-seed aggregates in the supplement / released JSON).

5 Limitations (domain shift)

- **No Indian phone-on-bag corpus yet.** All numbers use IRRI contact sensors plus ambient clean windows. The robustness ladder is a proxy, not a substitute for field labels.
- **Species coverage.** Pulse beetle and other pests are out of scope.
- **Reference comparison** to 84.51% is soft (protocol-matched claim, not identical reimplementation).
- **Single geography / grain type** in source data; storage bag materials and vernacular mics vary.

- **Screening aid only**—not legal or lab diagnosis; low-confidence UX must abstain.

6 Open release and outlook

Apache-2.0 code, distilled weights, bake-off tables, robustness/calibration JSON, and a static results dashboard are public.[1] Next: multi-seed CIs for the full advanced suite (released JSON when complete), tighter fusion gates, and—when feasible—phone-mic field collection with extension partners. Ground-up, small-model, offline tooling matches CCAI's call for resource-aware climate ML.

References

C. Balingbing et al. Application of a multi-layer CNN and acoustic device for stored rice pests. *Computers and Electronics in Agriculture*, 225:109297, 2024.

IGMRI. Annual Report. Indian Grain Storage Management & Research Institute, 2015.

World Bank / FAO reports on post-harvest loss (food loss & waste). See also regional grain storage extension literature.

[1] Anonymized for review; camera-ready will cite the repository.